

Book I.

Definitions

1.

A *point* is that which has no parts.

2.

A *line* is length without breadth.

3.

The extremities of a line are points.

4.

The straight or right line is that which lies evenly between its extremities.

5.

A *surface* is that which has length and breadth only.

6.

The extremities of a surface are lines.

7.

A *plane surface* is that which lies evenly between its extremities.

8.

A *plane angle* is the inclination of two lines to one another, in a plane, which meet together, but are not in the same direction.



9.

A *plane rectilinear angle* is the inclination of two straight lines to one another, which meet together, but are not in the same straight line.



10.

When one straight line standing on another straight line makes the adjacent angles equal, each of these angles is called a *right angle*, and each of these lines is said to be *perpendicular* to one another.



11.

An *obtuse angle* is an angle greater than a right angle.



12.

An *acute angle* is an angle less than a right angle.

13.

A *term* or *boundary* is the extremity of any thing.

14.

A *figure* is a surface enclosed on all sides by a line or lines.



15.

A *circle* is a plane figure, bounded by one continued line, called its circumference or periphery; and having a certain point within it, from which all straight lines drawn to its circumference are equal.

16.

This point (from which the equal lines are drawn) is called the *centre* of the circle.



17.

A *diameter* of a circle is a straight line drawn through the centre, terminated both ways in the circumference.



18.

A *semicircle* is the figure contained by the diameter, and the part of the circle cut off by the diameter.



19.

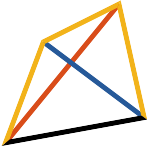
A *segment* of a circle is a figure contained by a straight line and the part of the circumference which it cuts off.

20.

A figure contained by straight lines only, is called a *rectilinear figure*.

21.

A *triangle* is a rectilinear figure included by three sides.



22.

A *quadrilateral figure* is one which is bounded by four sides. The straight lines  and  connecting the vertices of the opposite angles of a quadrilateral figure, are called its *diagonals*.

23.

A *polygon* is a rectilinear figure bounded by more than four sides.



24.

A triangle whose sides are equal, is said to be *equilateral*.



25.

A triangle which has only two sides equal is called an *isosceles triangle*.

26.

A *scalene triangle* is one which has no two sides equal.



27.

A *right angled triangle* is that which has a right angle.



28.

An *obtuse angled triangle* is that which has an obtuse angle.



29.

An *acute angled triangle* is that which has three acute angles.



30.

Of four-sided figures, a *square* is that which has all its sides equal, and all its angles right angles.



31.

A *rhombus* is that which has all its sides equal, but its angles are not right angles.



32.

An *oblong* is that which has all its angles right angles, but has not all its sides equal.



33.

A *rhomboid* is that which has its opposite sides equal to one another, but all its sides are not equal nor its angles right angles.

34.

All other quadrilateral figures are called *trapeziums*.



35.

Parallel straight lines are such as are in the same plane, and which being produced continually in both directions would never meet.